

Secure Boot: Ensuring Trust in IoT Devices

Security of IoT devices is critical as they are ubiquitous in home and industrial environments today. Secure boot is a security feature designed to ensure that only trusted software runs during a device's startup process.



It is also responsible for ensuring the proper operation of various hardware components within the device.

Given the integral nature of firmware in a device's functionality, its security is paramount. If an attacker gains the ability to manipulate or replace the firmware, they can potentially gain full control over the device, compromising both its operation and the data it processes. A notorious example of firmware vulnerability exploitation is the emergence of the Mirai botnet. This botnet infected thousands of IoT devices by exploiting weak or default credentials, enabling attackers to manipulate the devices' firmware. Once compromised, the devices were recruited into a botnet and used to carry out large-scale Distributed Denial of Service (DDoS) attacks. This case highlights the significant risks associated with unsecured firmware and the importance of implementing strong security measures to prevent such exploitation. Hence, secure boot is a critical security feature for IoT devices, ensuring that only trusted software runs on the device during its startup.

Low-end embedded devices are becoming increasingly popular due to their compact design and ease of use. These devices have seamlessly integrated into our daily lives, powering everything from smart home appliances to industrial IoT systems. However, their widespread adoption has also exposed them to significant security risks. One major concern is their vulnerability to malicious software injections. This is particularly critical given their limited user interface and lower levels of direct user oversight, which make detecting such threats challenging. To address this, the concept of secure boot has been introduced as a pivotal security measure for IoT devices.

Secure boot ensures that only trusted and authenticated software is allowed to run during the startup process. By leveraging cryptographic techniques, this mechanism validates

the integrity and authenticity of the firmware, safeguarding these devices against unauthorised modifications and malicious code. As the reliance on embedded devices continues to grow, implementing robust security measures like secure boot will play a vital role in ensuring their safe and reliable operation.

The role of firmware in IoT devices

Embedded devices rely on a bootloader code that plays a crucial role in the startup process by loading the firmware. Firmware is a specialised piece of software stored in the device's non-volatile memory, often referred to as read-only memory (ROM). This firmware enables the device to perform its intended functions and is composed of several key components, including the kernel, bootloader, filesystem, and additional resources.

Cryptographic signatures

For secure boot to be effective, a cryptographic signature must be embedded in the bootloader. Two common cryptographic methods used are RSA and ECC (elliptic curve cryptography). Both methods help verify the integrity and authenticity of the bootloader and firmware. Here's how these mechanisms work.

When a device manufacturer or trusted entity creates the bootloader (and